McKnight Foundation's Collaborative Crop Research Program (CCRP)
Eastern and Southern Africa Communities of Practice
GIS Getting started guide

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This GIS Getting started guide has been prepared to accompany the **GIS-data and starter CD for CCRP**. Its aim is to get you started in utilising the data provided on the CD by using QGIS software. This guide takes you through some simple exercises with step by step instructions. There is much more to learn in order to make full use of GIS and GIS data, but it is easier to explore on your own once you have learned the basics.

This guide assumes you have installed the software. It is provided on the CD and you can also download and install it directly from the QGIS web page at www.qgis.org (the exercises of this Guide were done by using QGIS 1.7.0). Copy the data folders Kenya and Mozambique to your hard drive. These two databases are used for the exercises below.

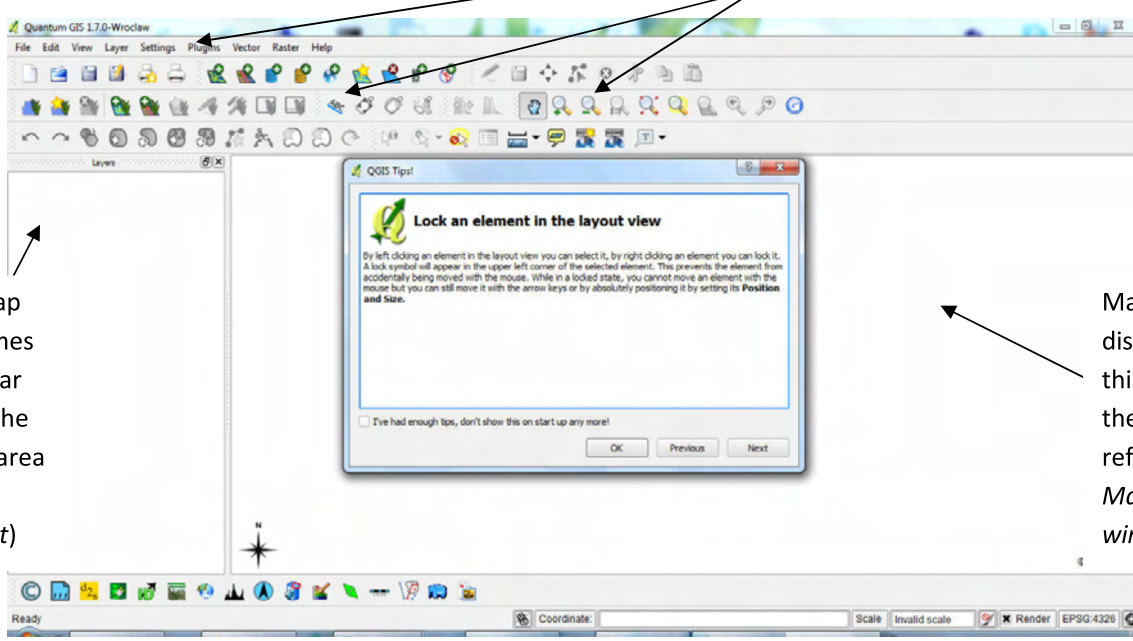
This guide and the exercises assume you have basic computing skills and are familiar with using Windows. The exercises build on each other, so you need to do them in the order they appear in this guide. A lot of details are included in the early exercises, but as you progress you will find less detail. You will need to remember what you learnt earlier and try things out.

As you work through this guide you will find terms that are not clear and concepts that are at first confusing. The best way to grasp them is by using them. For example, if the idea of a 'layer' is not clear, don't worry. Try to follow the exercises: soon you will be using layers, and shortly afterwards you'll understand what they mean!

Launch the software. It looks like the image below. The left section will contain a list of map layers you open (names of files), and the maps will be shown in the window on the right. Close the Tips from the red x button on the upper right corner and get started!

Main menu

Tools



List of map layer names will appear here (in the text this area is called *Layers list*)

Maps will be displayed in this window (in the text referred to as *Main map window)*

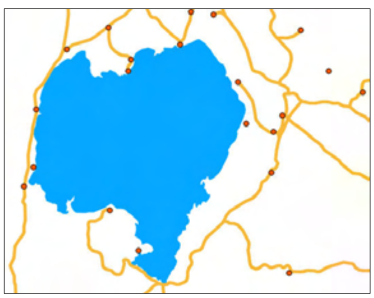
1. Display data and build a suitable legend in order to make a thematic map

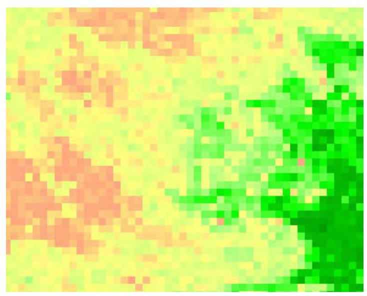
In this first exercise you open the annual climate map layer for Kenya. You also learn to make a legend in order to display the data.

The annual climate data file (named CLIann.shp) is an ESRI Shape file (.shp) which is a vector layer format. Vector layer data have features such as points, lines or polygons (areas). ESRI Shape file is the most common vector layer format used.

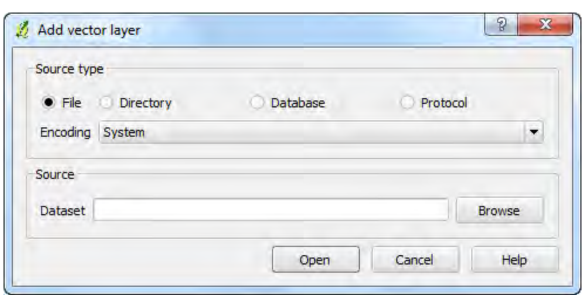
This particular annual climate map layer is exceptional because it looks like a grid or an image. Grids and images are raster data made out of rows and columns (a matrix) of squares called pixels, each with a data value. In this case, however, as the file has been saved as a Shape file, it is a vector layer and the little squares are polygons.

Below on the left is a vector map made out of a point layer (populated places), a line layer (roads) and a polygon layer (lakes). On the right is a sample of a raster map layer (tree cover). Each pixel on this raster map layer has a data value. Colours are assigned by categorising pixel values.





Open a vector map layer: Click  (Add Vector Layer) and a window below opens.

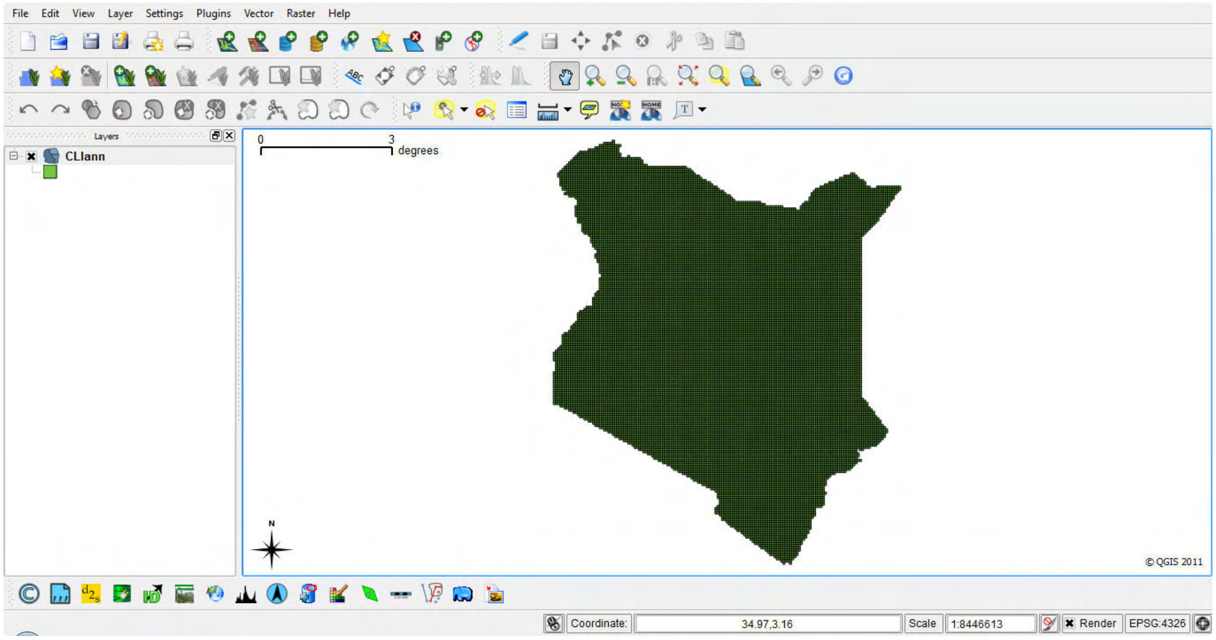


Keep the option File selected and Encoding System as it is.

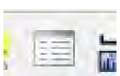
Click Browse and browse to the folder Kenya/Mudspring, which you copied to your computer, and select the file CLIann.shp,

Click Open, and again Open.

QGIS picks a colour randomly for each layer, and all polygons are displayed in the same colour (below it's green with a black outline) and thus no information can be seen.

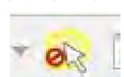


Look at the **attribute table** attached to this map layer to see what kind of climate data could potentially be displayed by this map layer.

 Click the table icon to open the attribute table.

See what the column names mean from the **metadata** (data about data) in the Kenya/Mudspring folder (in an Excel file).

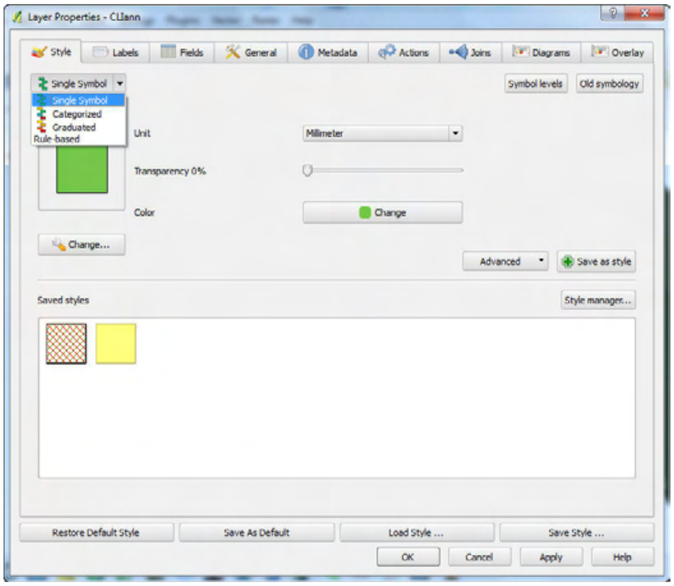
Each row in the table is attached to a square polygon on the map layer. You can confirm this by **selecting some lines** in the table and seeing how those square polygons get highlighted in yellow. Click row number 0 (click the actual number, not elsewhere on the line), hold the shift key down and scroll down in the table and click row number 510, the first 510 rows in the table get highlighted. See what happens on the map: the very northern parts of the country get highlighted.

To **deselect** the rows (get rid of highlighted rows in the table and map), go to the main map window (the main software interface) and click the icon Deselect features from all layers . Alternatively click the 'Unselect all' icon, at the bottom of the Attribute table window.

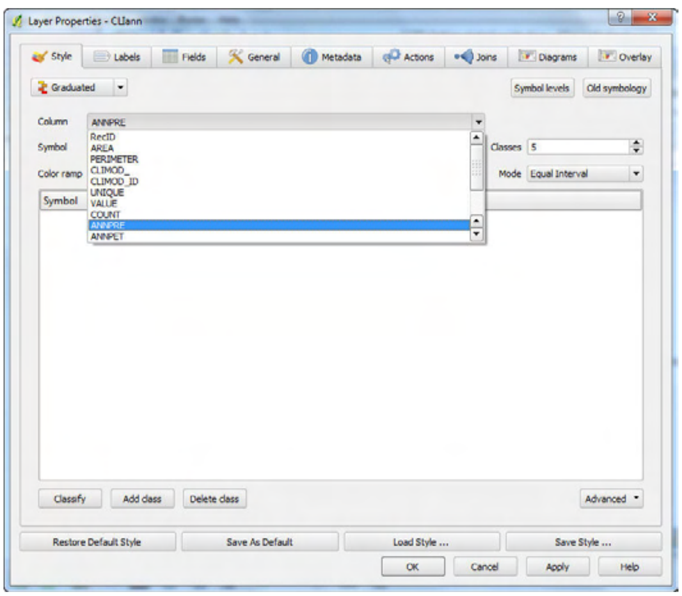
You would now like to see annual precipitation displayed. The data for this is in column ANNPRE.

To select the **desired data column** for the map, to **categorise the data** and to **select good colours for the legend**, do the following i.e. **change Layer properties**.

Double-click the layer name (CLIann) OR the legend colour box in the Layers list (the section on the left in the main software interface). The following Layer properties window opens.

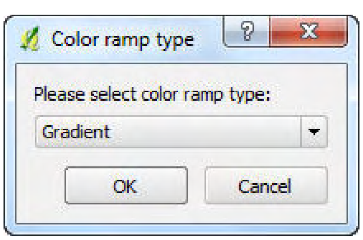


From the drop-down window, change the Single Symbol to Graduated.

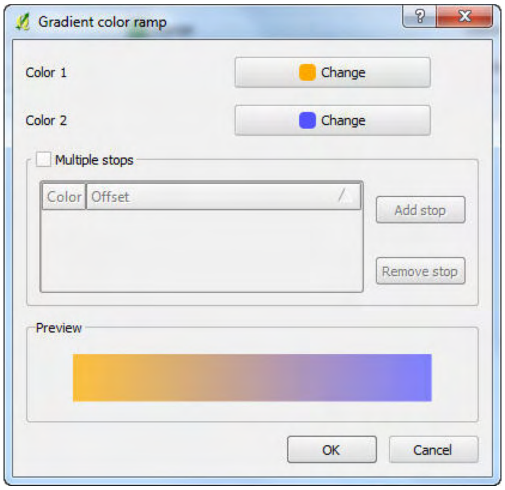


Select the Column you want the data to be picked as ANNPRE (i.e. annual precipitation). Select the number of Classes as 5.

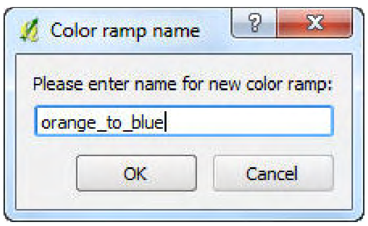
In the same window, click the Color ramp drop down menu and select New color ramp.



Select Gradient (you want to display annual precipitation as a continuous colour gradient). Click OK.



Click Change for Color 1 and select orange (low annual precipitation). Click Change for Color 2 and select blue (high annual precipitation). Click OK.



Give the new color ramp a name by typing, orange_to_blue. Click OK.

You are again back in Layer Properties window. Click Classify (lower left corner) in order to have the legend categories and their colours appear.

Click Apply and look how your map displays with this legend.

There are two problems to solve:

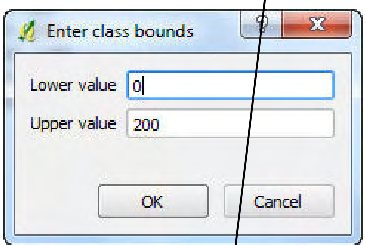
First, each little square has a black outline and thus the filling (orange to blue) does not show well. You need to make the outline the same colour as the filling.

Second, you need to know something about East Africa rainfall to be able to categorise it in a meaningful way. By default, the software uses Equal Interval (see Mode in Layer Properties window). But if you are making a map to your report, you want to see more rounded numbers in your categories (not e.g. 396.6000 – 793.2000), so you should categorise your data manually. If you don't know the subject area and don't know what category boundaries would be best, play around with the different Modes (by selecting Natural Breaks, Standard Deviation etc) and see what Mode picks the variation in a good way, i.e. Select a Mode, click Apply, look at your map. Select a different Mode, click Apply, look at your map.

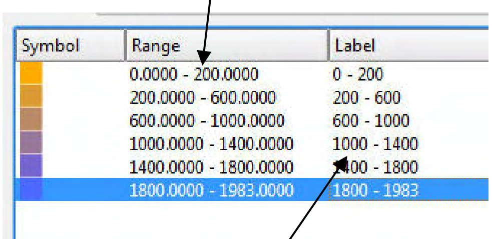
When you want a map for your report, make the categories manually to get rounded numbers as category boundaries. Let's say you would like to display Kenya annual rainfall by using the following 6 categories.

< 200
200-600
600-1000
1000-1400
1400-1800
>1800
In the Layer Properties window, change the number of classes to 6. Click Classify.

Double-click the number range under Range for the first category (smallest rainfall).

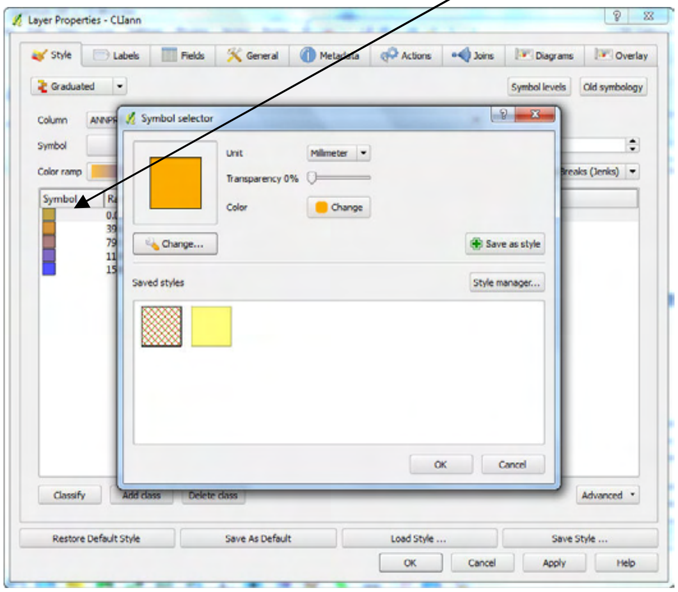


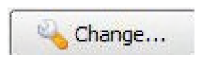
Set the range for the lowest rainfall category as Lower value 0 and Upper value 200. Click OK. Do all the categories the same way, picking the ranges from above (200-600, 600-1000 and so on). Click Apply to see how the map changes when it applies your new manually entered categories.

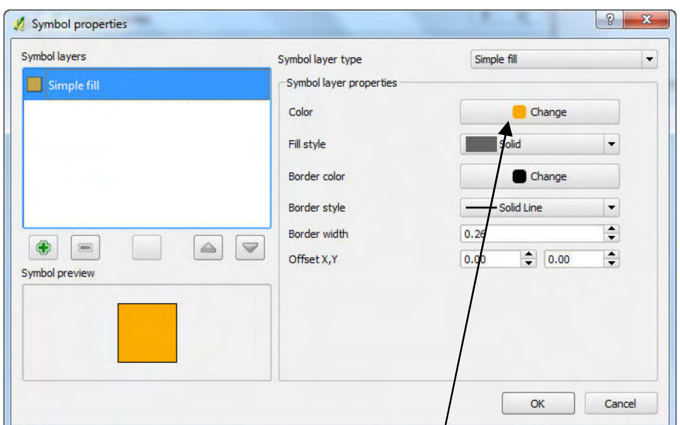


(Unfortunately the Labels in the legend do not get updated automatically according to your new categories. Change them manually by double clicking the range and typing the class bounds.)

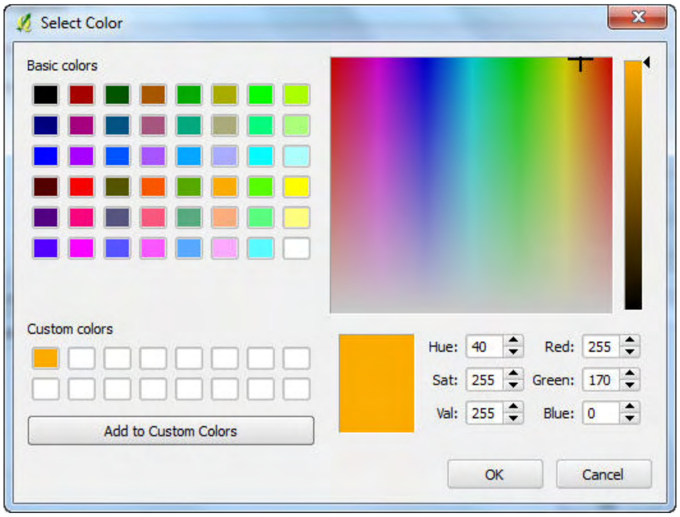
To get rid of the black outlines, Double-click the little orange box in the legend window (in the Symbol column left of Range and Label).



A Symbol selector opens. UNDER the orange box, click Change .



In Symbol properties window, click Change for Color.

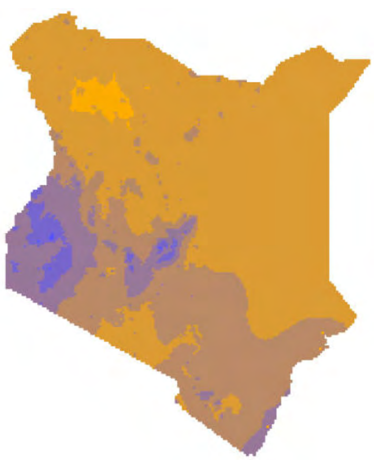


Click Add to Custom Colors (so that the particular hue of orange can be picked later as an outline colour). Click OK.

Click Change for Border color. Pick the orange from the Custom colors. Click OK to all.

Do the same for each colour in the legend up to blue.

At the end click Apply. Your map should look like this.



Before closing the Layer Properties window, click **Save style** (if you have closed it, double click the file name or the legend on the Layers list to open Layer Properties, click Save style). This saves your categories and colours for future use.

Give your style file a name, e.g. Cliann_legend.qlm and save it in the same folder where your CLIann.shp for Kenya is (Mudsprings/Kenya). When you display the annual precipitation file again, you can just click Load style, browse to Cliann_legend.qlm and you don’t need to build a legend again for this layer.

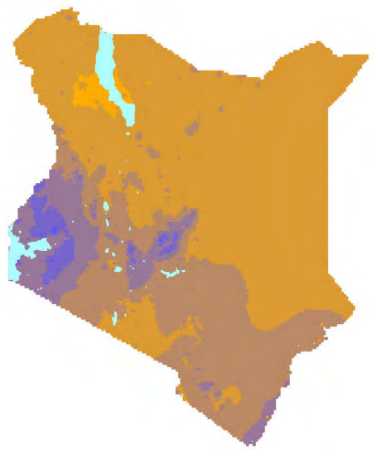
Now display lakes on the same map.

Open a layer called HYDdnnetp.shp. Follow the instruction at the beginning of this exercise. As before, it opens with a random colour.

To change Layer properties (the style to display it) double-click the layer name or the little legend box in the Layers list to open Layer Properties window.

Keep Single symbol selected. Click Change under the coloured square 

. Change the Color into light blue/turquoise. Click OK. Click Border colour and select the same light blue/turquoise. Click Ok to all.



Before Exiting QGIS (File/Exit) or removing the layers, **save your project so that you can return to it later**. Use File/Save Project As, browse to the desired folder, give your project a name annual_precip.qgs and click Save.

When you return back to the work done in this first exercise, launch QGIS and select File/Open Project, select the file you just saved (annual_precip.qgs) and everything opens just the way you left it (this function saves the order of layers and their legend settings).

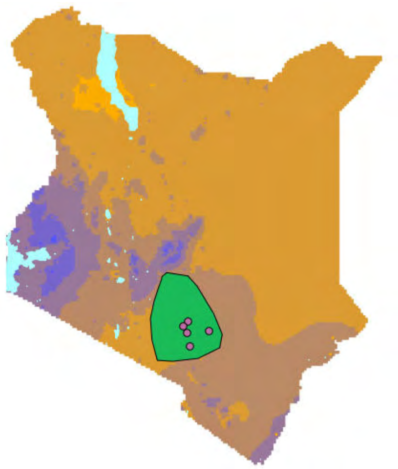
2. Display data compiled in Google Earth in QGIS and use labelling

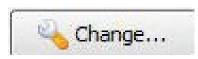
In this exercise you use two files (based on an imaginary project area in Southern Kenya) that have been created in Google Earth. You can follow the instructions on how to create them in the other guide called ***Capturing project areas in Google Earth****. However, the files sorghum project areas.kml and sorghum project locations.kml have also been save in the folder Kenya/Example.*

You will open these files on top of the previous project annual_precip.qgs (with the annual precipitation and lakes) and display labels for the sorghum project locations file.

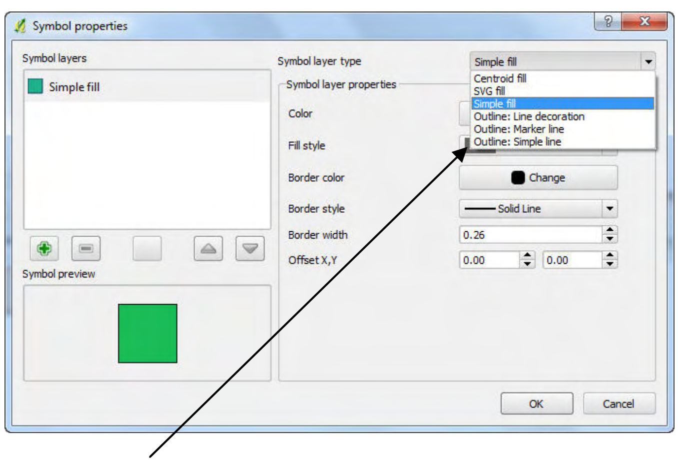
Open the Project file annual_precip.qgs that you saved in the first exercise (File/Open Project). Add the two vector layers sorghum project areas.kml and sorghum project locations.kml.

Your map looks like this:

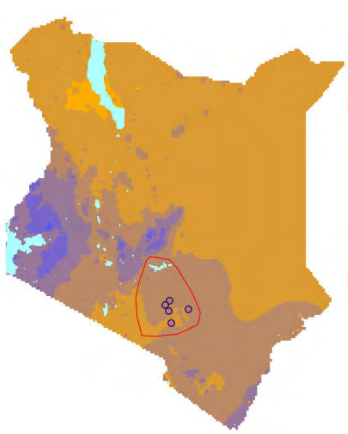


You would like to make the sorghum project area transparent so that you can see the rainfall data through it. Change the Layer properties: Double-click the layer name sorghum project areas or its legend colour box (in the Layers list). The Layer Properties window opens. Click Change 

to modify the Symbol properties:

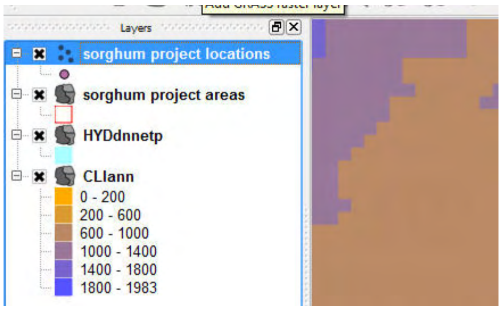


Select Outline: Simple line for the Symbol layer properties. Click Color. Click red colour in the palette. Click OK to all. Your map looks like this:

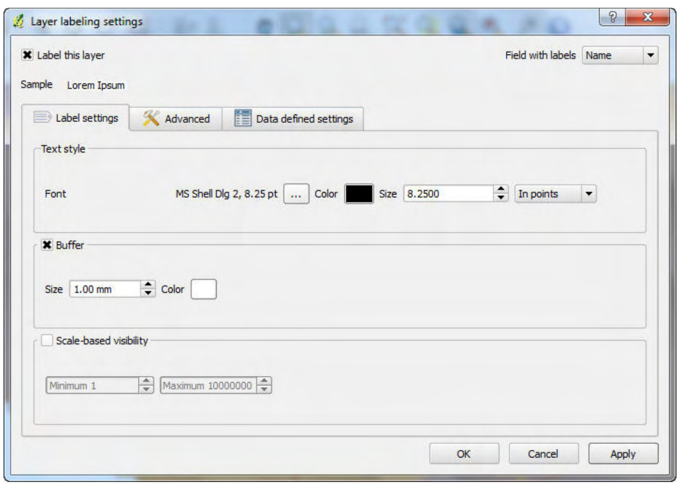


Zoom to your project area by picking the Zoom in tool and drawing a box around the project area. 

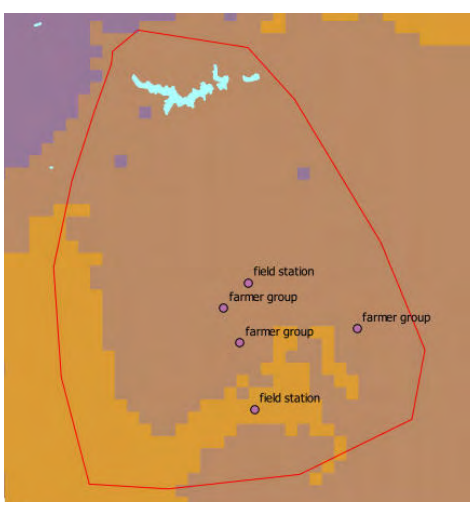
Put the labels on for the project locations: Highlight file name sorghum project locations in the Layers list.



Click the icon for Labeling  (There is an option to work on Labels also through Layer Properties)



In the Layer labelling settings, tick the box Label this layer, for Field with labels select Name, untick Buffer (white area around text), click Advanced tab, adjust the Label distance so that the label is not too close and not too far from the point (probably about 2, check by Apply). Click OK. Your map looks like this:



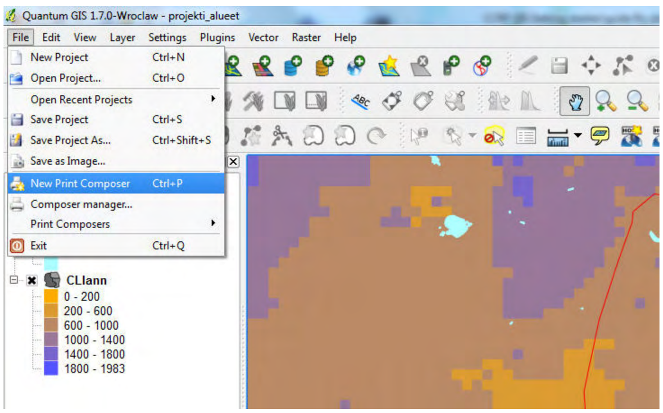
Save your project. Use File/Save Project As, browse to the desired folder, give your project a name annual_precip_areas.qgs and click Save.

3. Print the map

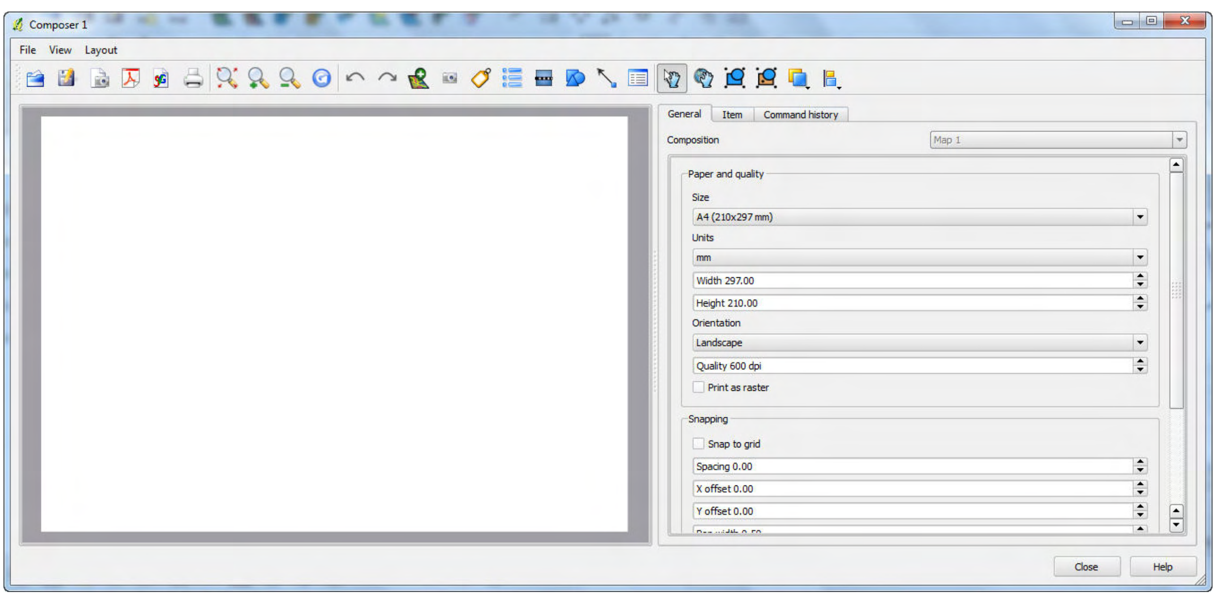
In this exercise you print the map of the previous exercise (annual precipitation in your project area). To be able to print a map from QGIS, you need to ***make a layout*** *(compose it), save that layout as an image, and print that image.*

Open the project that you saved in the previous exercise annual_precip_areas.qgs.

Open **Print composer** from File/New Print Composer.



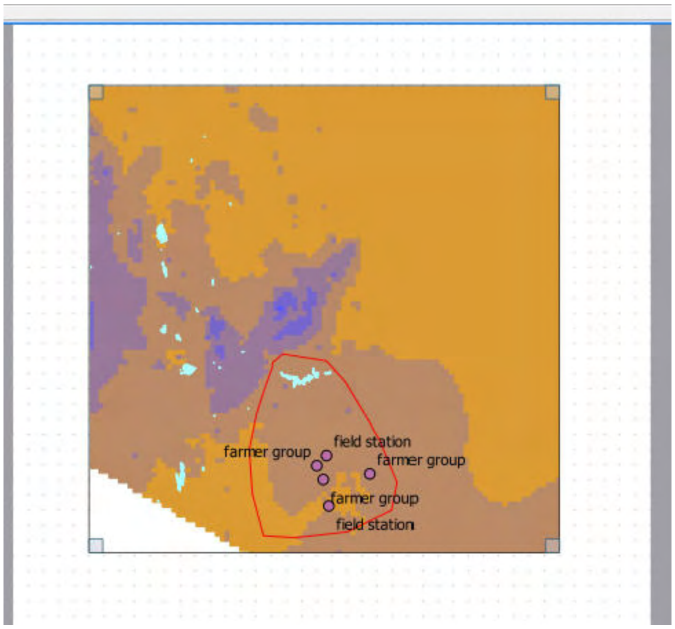
The following window opens.



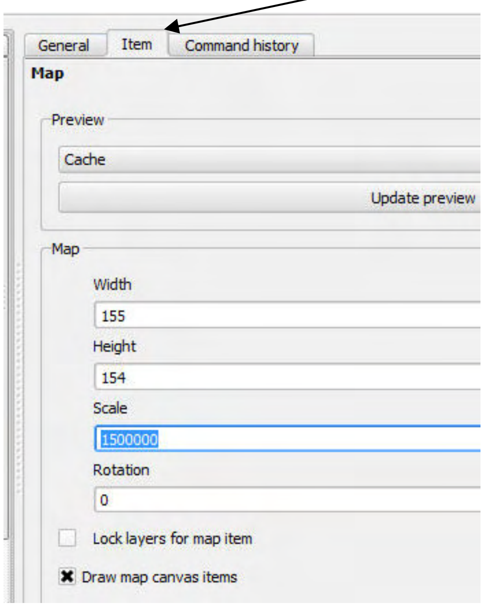
Keep the Size as A4. Turn the Orientation to Portrait. Quality 300dpi is enough as this map is not going to be used in a publication (assume it is just a working copy).

Check Snap to grid (helps you align the components of the map). Set Spacing to 5, Pen width to 0.25, Grid color to grey, and Grid style to Dots.

Pick a tool called Add new map . Draw a box for the map on the A4 paper.



The map does not scale automatically. Select the Item tab and change the scale manually to about 1500000.

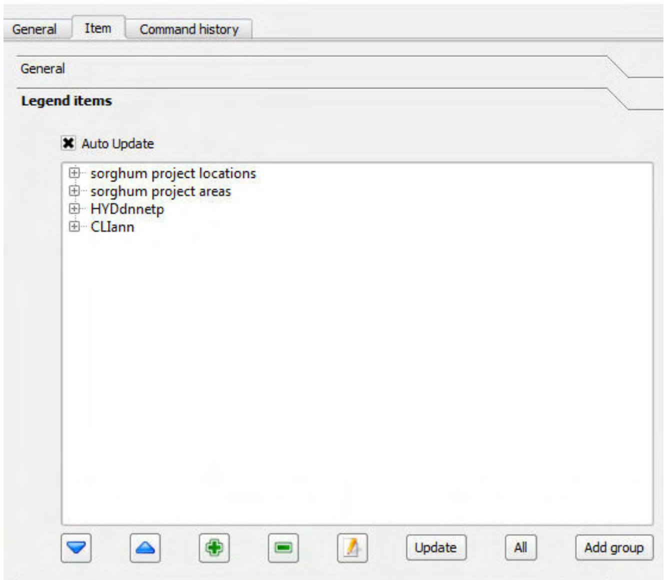


Use the Move item content tool  for panning the map and centering it to the middle of the window (click on the map and while you hold down your mouse button, drag and release).

Expose the empty lower part of your A4 paper. You are going to put the legend there.

Pick the Add new vect legend tool  and click on the layout (A4) where you would like the legend to appear.

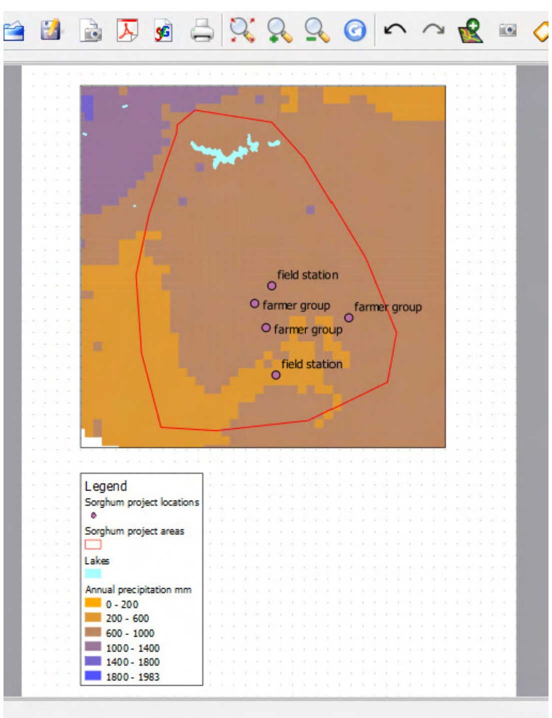
In the window on the right you can manipulate the legend.



Click HYDdnnetp to highlight it and click the ‘pen and paper tool’. Change the name to Lakes.

Do the same for CLIann and change it to Annual precipitation mm.

Do the same to both project layers and start them with capital letter.



Maps in reports and publications should have more things than this, e.g. a scale bar, coordinates around the map window that show the geographic location of the area, index map of the whole country, maybe north arrow. You will learn to make a publishable map in the last exercise, not here.

Save your map by File/Export as Image...

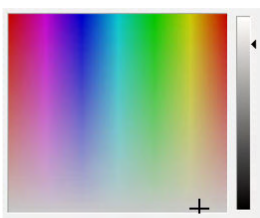
Save it as a jpg (e.g. tif images are very large files). Give it a name. Open it in some image viewing software and print.

4. Make a selection in order to create a new layer with selected data only

In this exercise you will make a little black and white map of Kenya in which you show the country boundary, lakes, few big towns with labels (you need to select these few towns), and your project area.

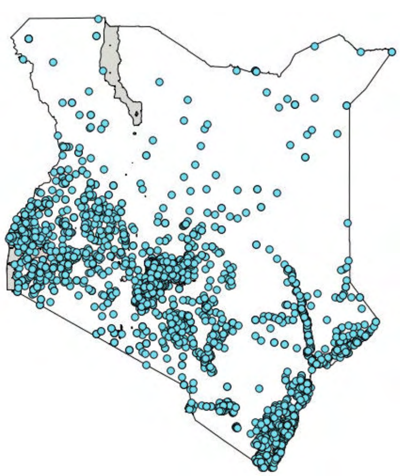
From the Kenya/Mudspring folder, add vector layer called POLbdy.shp (country boundary). Get rid of the fill colour and have it display with a black outline only (Follow earlier instruction on how to change Layer properties).

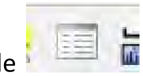
Open lakes vector layer HYDdnnetp.shp. Change the style so that lakes display by light grey. Use the flexible palette and toggle the grey scale bar.

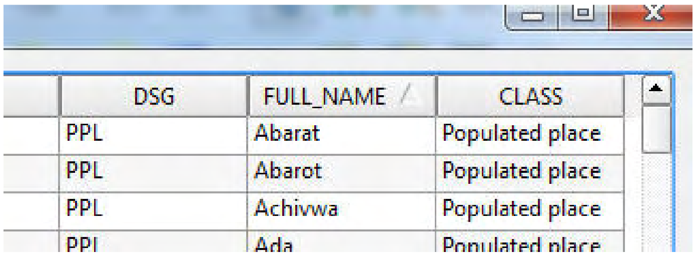


Add vector layer populated places DEMpppoint.shp.

Much too many populated places in Kenya! You need to select the bigger towns only.



To select only few towns from the hundreds, Click open the attribute table  and click FULL_NAME to organise the place names in alphabetical order. (You need to have the correct layer highlighted in the Layers list in order to get the correct table open).



DSG	FULL_NAME	CLASS
PPL	Abarat	Populated place
PPL	Abarot	Populated place
PPL	Achivwa	Populated place
PPL	Ada	Populated place

By holding ctrl key down, pick the few towns you want (Embu, Garissa, Kakamega, Kisumu, Mombasa, Nairobi, Nakuru, Nyeri). Don't click on the name, click on the row number so that the whole row gets highlighted. (Leave the table open without clicking anywhere on it).

Go to the main QGIS interface (where the map window and Layers list are) and right-click on the layer name DEMpppoint in the Layers list, select **Save selection as...**

In the window that opens (Save vector layer as) Browse for Save as and give this new layer a name selected_towns.sph and click Save. Click OK.

Open the new layer selected_towns.shp.

Remove the layer DEMpppoint (the layer with hundreds of towns) by right-clicking on layer name and select Remove.

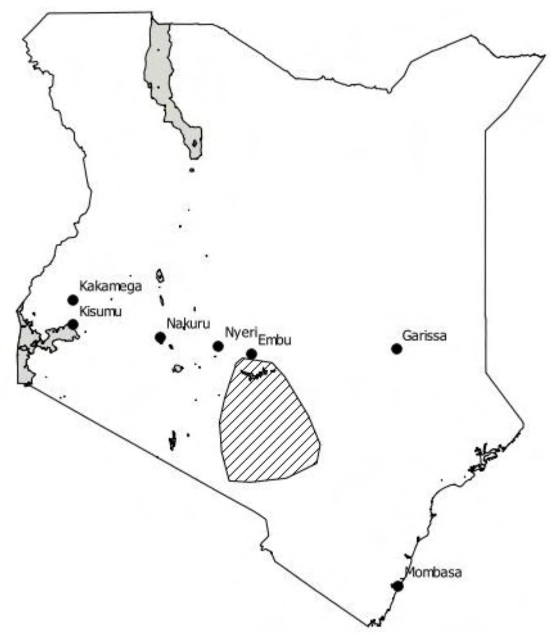
Change the type of point symbol to a black round dot. (Double-click the layer name selected_towns to get to Layer Properties. Refer to earlier instructions (section 2) to change styles)

Put labels on for the selected_towns map layer. (Refer to earlier instructions on how to work on labels)

Open vector file sorghum project areas.kml. (Kenya/Example)

Change the style in the following way. Double-click the layer name sorghum project areas to get to Layer Properties. Click Change under the coloured square. Change Color to black. **Change Fill style BDiagonal**. Click OK and OK.

Your map should look like this:



Save your project as map_for_report.qgs as we will pick this project for the last exercise in this guide.

5. Use data with different projections by using the QGIS 'on the fly' projection utility

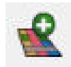
In this exercise you experiment with data in different coordinate systems. This is important as maps in different projection systems do not display on top of each other correctly. To solve the problem you use the unique QGIS 'on the fly' projection utility (many popular commercial GIS software do not have this capability). You will use a raster layer that was created in a metric system (coordinates are in meters), in Mollweide projection with central meridian at 0.00, and a vector country boundary that is in geographic (latitude, longitude degrees) coordinate system.

Before starting this exercise Exit QGIS and launch it again. This is to make sure you get rid of any project specific projection settings that was picked from the data that you used just before starting this exercise.

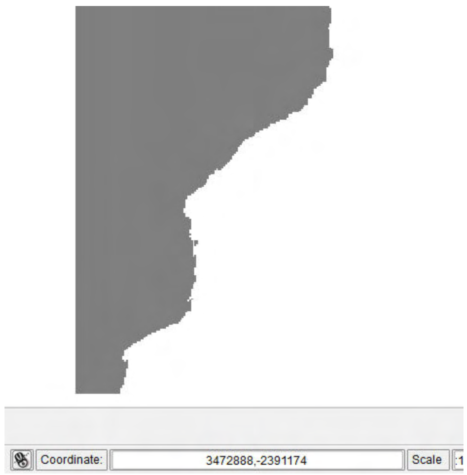
Most of your data is in the Geographic coordinate system (latitude, longitude) with the WGS84 (World Geographic System) datum. This is also the default Coordinate Reference System (CRS) QGIS uses if you don't change anything.

Check what CRS you have selected in your software and what will be used as the **default CRS system for all new projects that you start**: click Settings on the main menu, then click on Project properties. Click the tab labelled Coordinate Reference System (CRS) to display all the datum options. Scroll down the *Geographic Coordinate Systems* list till WGS84 appears, and select it. Click OK.

Now go again to Settings on the main menu. Click on Options. With the CRS tab selected look at the Default Coordinate Reference System for new projects. You should have here the option EPSG:4326 – WGS 84 selected too. Do NOT yet check the Enable 'on the fly' reprojection by default. We'll do it later, after experimenting with one data layer that is not in your selected system. Select Use project CRS in the lower part of the Options window. Click OK.

Open a raster map layer  from Mozambique/Net_prim_prod/net_prim_prod81-03. This file is an Erdas .img file.

Your map looks grey and white. If you move your mouse over the map, you can see a metric X and Y coordinates moving in the in the little Coordinate window below the map window.

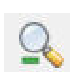


If you try to open some of the Mudspring data on this raster, the two layers would not show on top of each other. Try opening Mozambique country boundary Mozambique/Mudspring/POLbdy.shp. The map layer name appears in the Layers list, but the map does not show. Where did it go?

Just for curiosity, look for it by highlighting the layer name POLbdy.shp on the Layers list and clicking by Zoom to layer .

The Mozambique country boundary appears to the map window but the grey raster disappears. The country boundary is in its original latitude longitude degree system (move your mouse over the map and look at the little Coordinate window below the main map window).

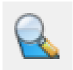
Again, just for curiosity, try to look where the two layers are in relation to each other.

Pick the tool  and click by it IN THE MIDDLE OF the Mozambique country boundary map as many times as needed to start seeing the raster layer appearing from the right bottom corner.

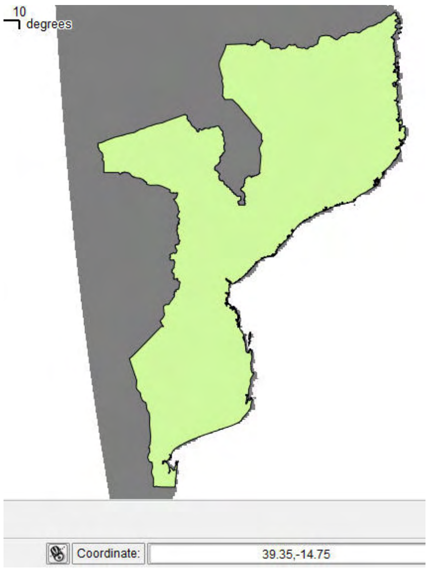
So **what to do to display data in different coordinate systems on top of each other**?

Go to Settings in the main menu again. Pick Options and check ON the **Enable 'on the fly' reprojection**...

Go to Settings. Pick the Project properties and check ON the box **Enable 'on the fly' CRS transformation**.

Now Zoom to either layer (i.e. either layer name highlighted in the Layers list) by clicking Zoom to layer .

You should see the two layers aligned. The coordinate system of your map window is now Geographic (lat, long, WGS84). You should also notice how the **raster has become warped**. Its originally straight left side is now curved. Once you create a legend for the layer (as described next) and can see the pixels, you'll see that they are not in straight rows and columns.

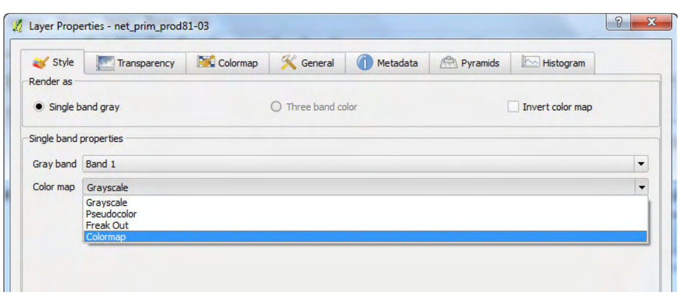


How to make the raster map display its data. As you can see from the metadata, the raster you have opened displays the change in biomass production between 1981 and 2003. Negative figures mean that biomass production has declined and positive figures mean that it has increased (kg/ha/year... see metadata for more details).

First, make your POLbdy layer show with border line only. (see exercise 2 how to do it) so that the raster can be seen through it.

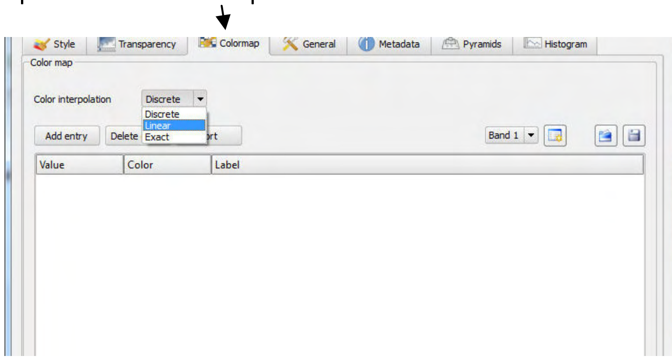
Start building a colour map by which to display the raster:

Double-click the raster layer name net_prim_prod81-03 to open Layer Properties window.



In the Style tab click the options for Color map. Select Colormap.

Open the tab Colormap.



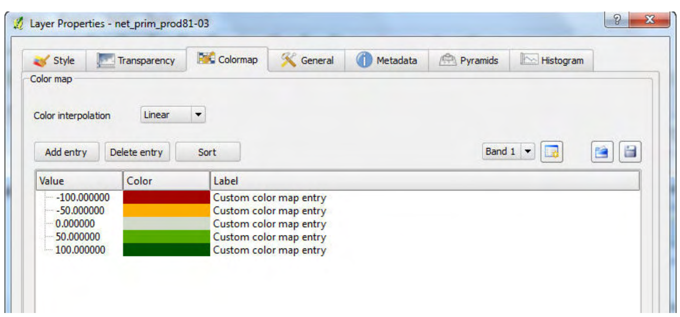
For Color interpolation select Linear (because you want a smooth transition between the colours).

Click Add entry. A new line appears for the legend. You might think about displaying this data as follows:

-100 (means biomass production has gone up by 100kg/ha/y, pick a red colour)
-50 (pick an orange)
-25 (pick yellow)
0 (no change, pick grey)
25 (biomass production increased by 25kg/ha/y, pick light green)
50 (pick middle green)
100 (pick dark green)

Double click the 0.0 and type -100. Double click the suggested pink colour and change it into dark red.

Click Add entry. So the same for each class as above until your colourmap looks like this:



Click Apply.

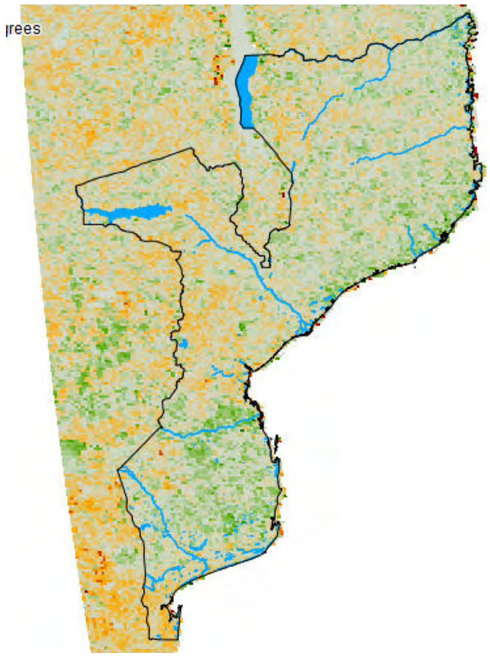
Click Save Style. Browse to the Mozambique/Net_prim_prod/ folder and save your style file as npp_red_or_grey_green.qml

Click OK to your Layer Properties window.

Open lakes layer on top of your raster. Mozambique/Diva_GIS_data/MOZ_water_areas_dcw.shp

Change lakes to blue (both fill colour and borders).

You might want to **change the order of layers** so that the boundary layer is on the top (one river hides the northern boundary). Click on the MOZ_water_areas_dcw in the Layers list and hold your mouse button down. Drag the layer downwards till you see a horizontal line between the POLbdy and net_prim_prod81-03 layers. Release the mouse and your water areas layer should now be between the two other layers.



If you want to return back to this work, save your project.

6. Make a printable map suitable for a project report (and build a legend for the tree cover raster)

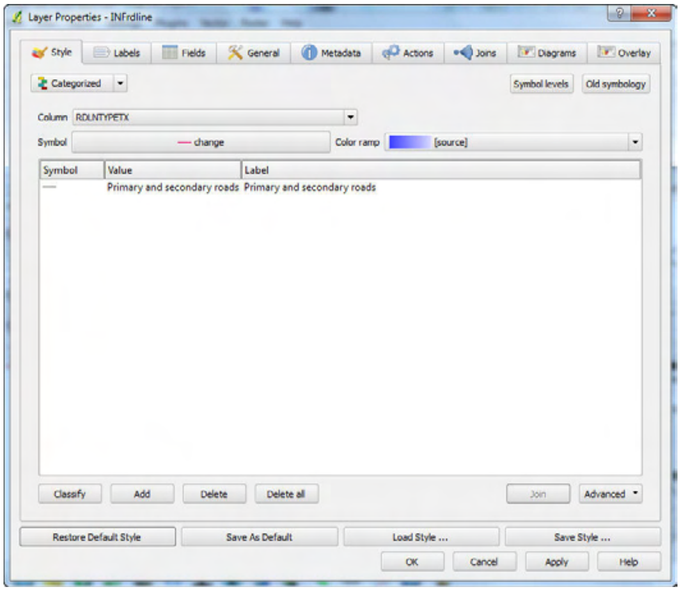
Maps in published work normally have ***coordinates around the edge of the map****. If they show a small area, they display* ***an index map*** *showing where the small area is located in the country. Maps also always have* ***a legend*** *so that the information in the map can be interpreted. They also have* ***a scale bar****. In this exercise you will make an image that is suitable for published work. Your map will show tree cover within the project area.*

Open the project that you saved at the end of exercise 3, i.e map_for_report.qgs (the black and white map of Kenya).

Zoom tightly to the project area, by selecting the Zoom In tool and dragging over the area you want.

Add few more layers:

1. Open road layer Kenya/Mudspring/INFrdline.shp. Display it in the following way: Open the Layer properties window (by double-clicking the layer name).

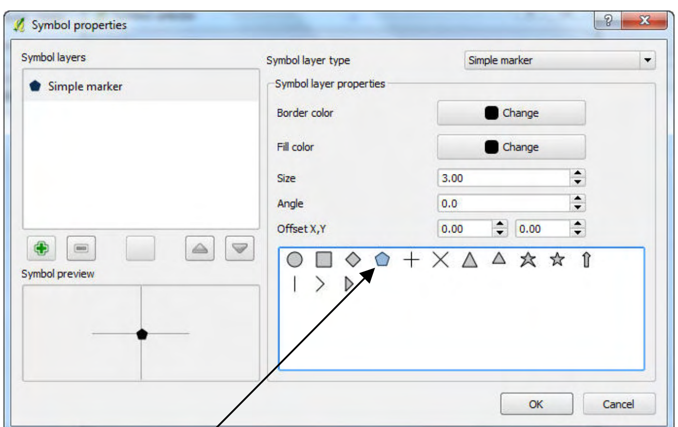


Under Style tab, select Categorized. For Column select RDLNTYPETX. Click Classify. Double-click on the Symbol for Primary and secondary roads. Pick 0.26mm black line for the Primary and secondary roads.

Delete the other two categories of roads (highlight on the list, click Delete).

2. Open sorghum project locations (Kenya/Example). Display it in the following way: Open the Layer properties window. Under the Style tab, select Categorized. For Column select Name. Click Classify. Double click the Symbol for the field stations, click Change 

.



Pick the pentagon and increase its size to 3. Click OK.

Double-click the Symbol for farmer group. Make the symbol a white dot with black edge, Size 3. Click OK.

Put the labels on for this layer. Make text size 12 or 13.

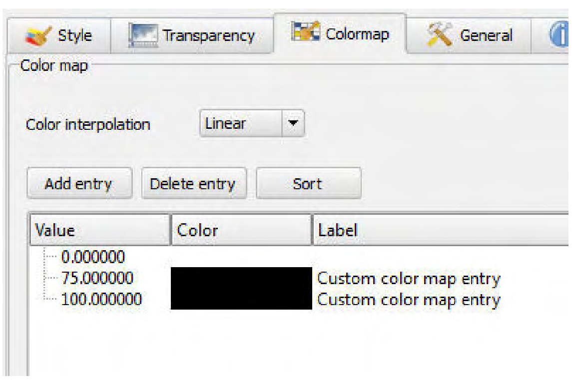
3. Open a raster layer Kenya/Modis_tree_cover/tree-cover. Drag it to the bottom of your Layers list (so that all the vector layers are on top of it).

In order to display data on tree cover layer, double-click the tree_cover layer name to get to Layer Properties. The first thing to try would be to select Color map as Grayscale with the Custom min /max values (Min 0 and Max 100), with Current Stretch To MinMax. But the map is still too dark for publication (and the grayscale is reversed, dark means sparse tree cover, light grey means dense tree cover... the latter is easy to reverse by putting Min 100 and Max 0, but the resulting map would be confusing).

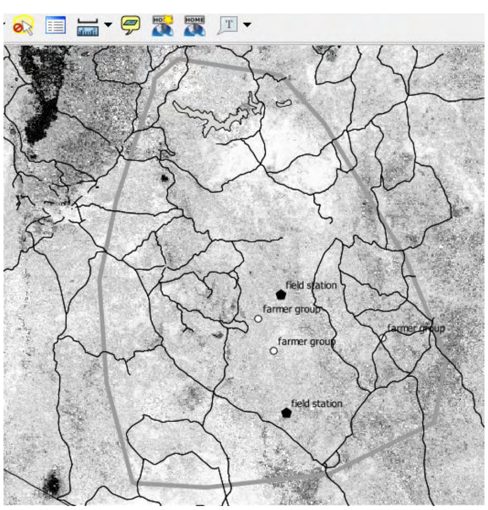
So, it is better to make a Colourmap as done before for the biomass data, but now you want it b&w.

To help you decide what kind of colour map you should be making for your specific small area, use the **Identify Features tool**  to see what kind of forest cover values your raster layer has. Select the tree_cover layer. Then select the tool and click with it all over your project area. The pixel value that appears in the Identify Results box is a percentage of tree cover. It appears that the maximum tree cover in the area is about 60%.

Make a black and white colour map. For this particular area (where the highest tree cover is about 60) it looks good if 75% is already black (but make an additional entry for 100%, otherwise areas with higher than 75% tree cover remain white). So make it like in the window below, and click Apply.



Now the roads and the border around the sorghum project area are both displayed as the same thin black line. That's not ideal. Change the border colour for the sorghum project area into grey and with a slightly thicker line. Your map should now look something like this:



Now start making the layout for your report image.

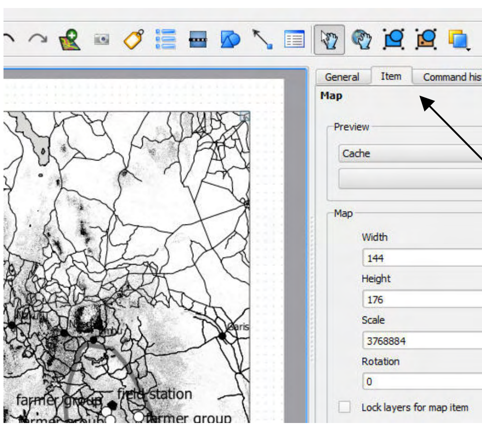
Click File/New print composer.

Make sure your paper size is A4. You might change the Quality to 600dpi (dots per inch) in order to make a very crisp (sharp, high resolution) image. To get all items (even small text) to show with high resolution, it is good to make maps like this initially to fit the whole A4 and reduce its size later according to the space available in a report or a publication. Remember to set the text font size large enough, so that you can reduce the size of the whole image later and your text will still remain readable.

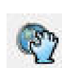
Snapping to grid is a good idea as you can align the content of your layout with each other. Adjust the Snap to grid anything between 3-5 and change the Pen width to 0.25 and Grid colour to light grey.

Now pick a tool called Add new map . Draw a box (by dragging over the area) for the main project area map on the right of the A4. Leave space for the index map and the legend on the left.

QGIS does not scale the map into your layout map window accurately according to the zooming, so you need to specify a suitable scale yourself.

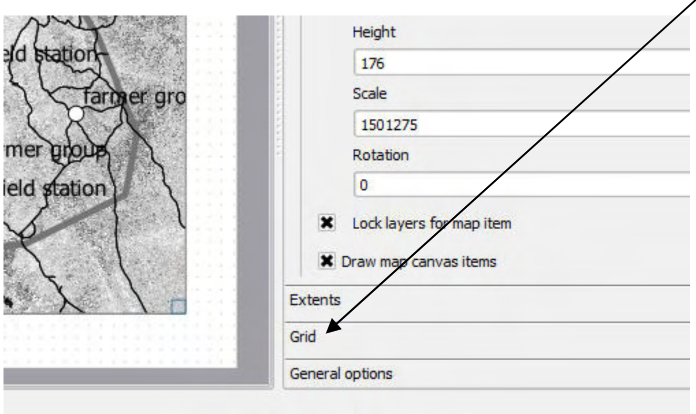


Pick the Select/Move item tool  and click with it on your map if it is not selected (it is automatically selected after drawing the box unless you click somewhere else). Go to the Item tab. In this tab, adjust the Scale of your map so that the project area fits closely to the window (something like 1500000 seems good if your window reaches close to the edges of the A4 area).

Use the Move item content tool  for panning the map and centering it to the middle of the window (click on the map and while you hold down your mouse button, drag and release).

Check the box Lock layers for the map item. This makes sure that when you start making your index map and reduce some of the layers for it, this ready map would stay the same.

Now make the coordinates show around your map. Click the Grid tab.



Check the box Show grid.

Select Grid type as Cross.

Put the Interval for X and Y as 1 degree.

Cross width 3, Line width 1.

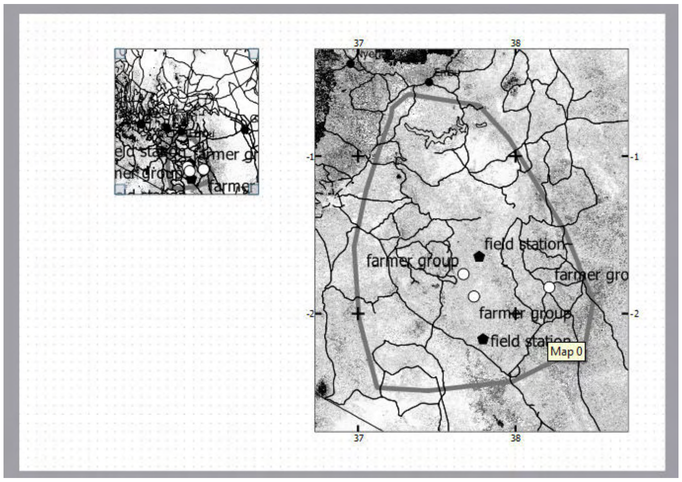
Line color Black.

Check the box Draw annotation (meaning write the degree figures around the map).

Select Annotation position as Outside frame. Annotation direction Horizontal.

Increase Font size to 12, Distance to frame either 1 or 2, Coordinate precision 0 (no decimals).

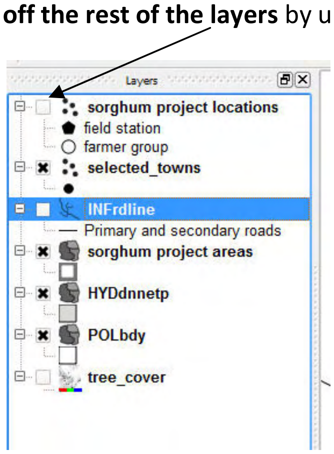
Now Add new map for the index showing the whole of Kenya. Make it smaller than the project area map and move it to the upper left corner.



Select your new map by  (it is automatically selected after drawing the box unless you click somewhere else).

In the Item tab, adjust the Scale to about 17000000 (if your window looks like the layout above).

An index map should have only country boundary, selected towns, lakes and the project area shown. **Click off the rest of the layers** by unchecking the boxes. Go to the Layers list.



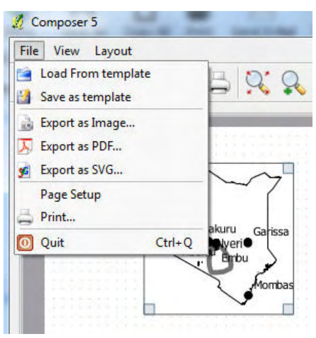
Go back to the Composer window and click Update preview. Check the box Lock layers for map item.

If you need, you can still go and adjust legends and label text size etc in the main map window.

Go to Grid again. Follow instruction above except put 4 or 5 degrees to Interval X and Interval Y, cross width 2 and Line width 0.5.

At this point **save your layout as a template** (in case you mess it up later).

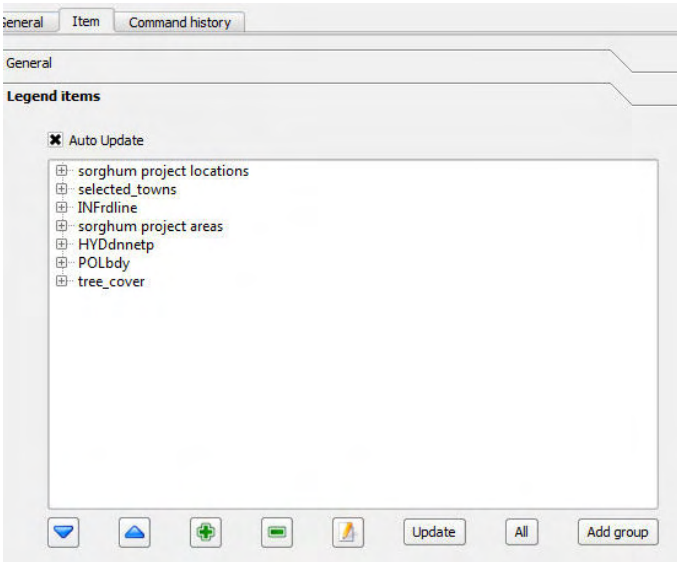
Take File/Save as template. Give it a name...



Now make a legend. Put on all the layers of the big map (check the boxes in the main Layers list). The legend tool picks only those layers that are displayed.

Pick the Add new vect legend tool  and click on the layout (A4) where you would like the legend to appear.

In the window on the right you can manipulate the legend, e.g. you don't need all the items described.

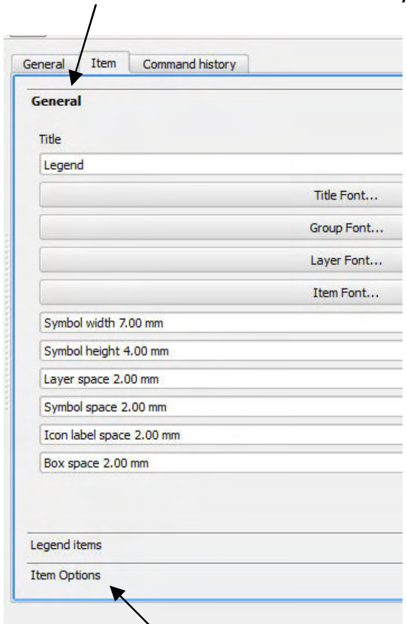


Delete selected towns, HYDdnnetp, POLbdy and tree cover (the latter because it is not a vector and this tool cannot make a legend for it). Use the green minus button below the list to delete items.

Edit the text in the legend by selecting an item in the legend and clicking the tool . E.g. start sorghum by capital letter, replace INFrdline by Roads.

By the blue arrows, move both Sorghum project related items up and leave the Roads last (as it is your project information that interests in this map, and Roads are just a reference).

In General sub-tab of the Items tab you can change the font size and the text for the legend.

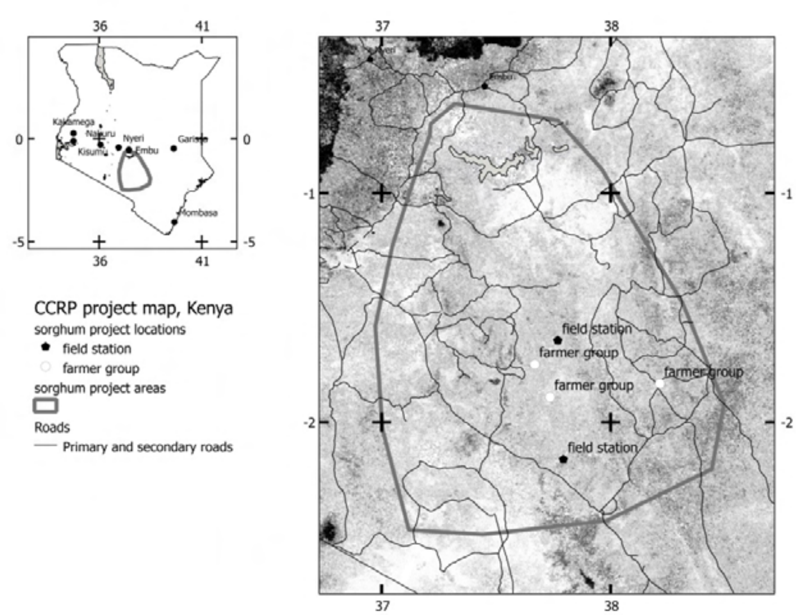


You can click Item Options on the bottom of the window and change e.g. the Frame colour etc. Uncheck the box Show frame to get rid of the frame altogether.

Now save your template. Also save your project.

Save your layout as an image, File/Export as Image... select the File type as jpg.

Look at your image in some image viewing software. Your map should now look something like this when it is Exported as Image.

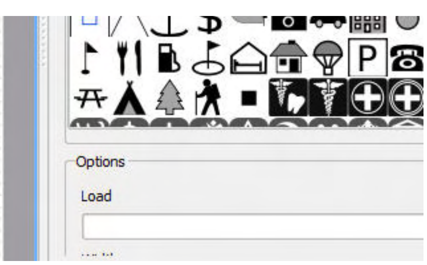


From here onwards things get a bit harder as there are no simple tools to do the rest.

There is no tool for **making a legend for the tree cover raster layer**, as the QGIS raster legend tool is still under construction.

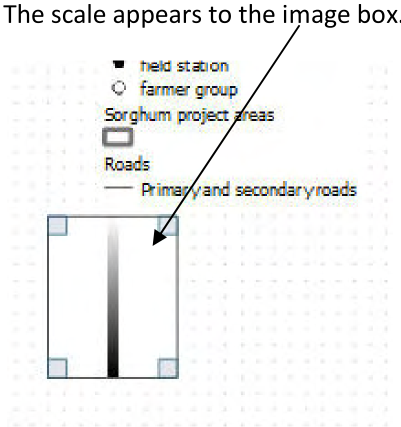
If you are an advanced computer user here is a hint on how to make it. Go to the Layers list. Open Layer Properties window to any of the vector map layers. Go to the colour palette. There is a useful grey scale bar in this palette. Take a print screen image, cut this bar out, and save it as an image.

In the Print Composer, pick a tool Add image . Click with it on your layout where you would like your grey scale bar to be.



Select your grey scale image file to Load.

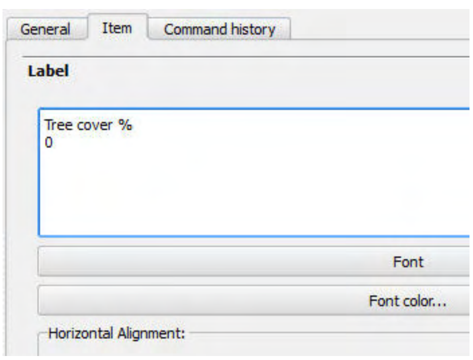
The scale appears to the image box.

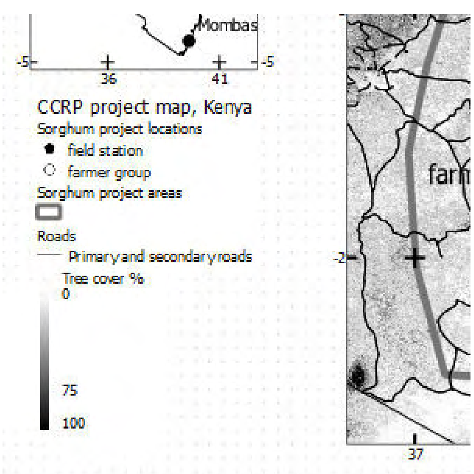


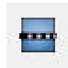
Get rid of the frame around it in the Items tab, General options sub-tab (uncheck the box Show frame).

Now make a text description to the bar by picking a tool Add new label  and clicking by it on the right side of the grey scale bar. Adjust location.

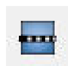
Change Label title to Tree cover % and hit enter to go to next row of text, write 0, hit enter so many times as to put 75 to the right place where the bar changes to black, add 100 lowest.

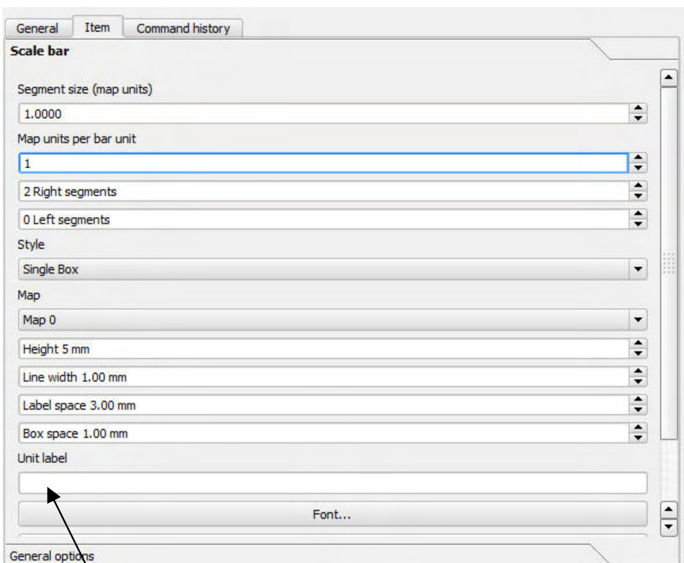




The QGIS Print Composer **scale bar** tool  is inadequate as it does not pick meters or kilometres as map units when your map layers are in degrees (even though the Measure line tool  works in kilometres). Your scale bar will come in degrees – which is not appropriate. However, it is possible to play around manually and get an approximately correct scale bar.

Go to your main map window and pick the **Measure line tool** . **Measure how long in kilometres** is one degree by measuring the scale bar of your main map window. It is 111km in this map location.

Pick the Add new scale bar tool . Click it on the layout where you would like the scale bar appear.

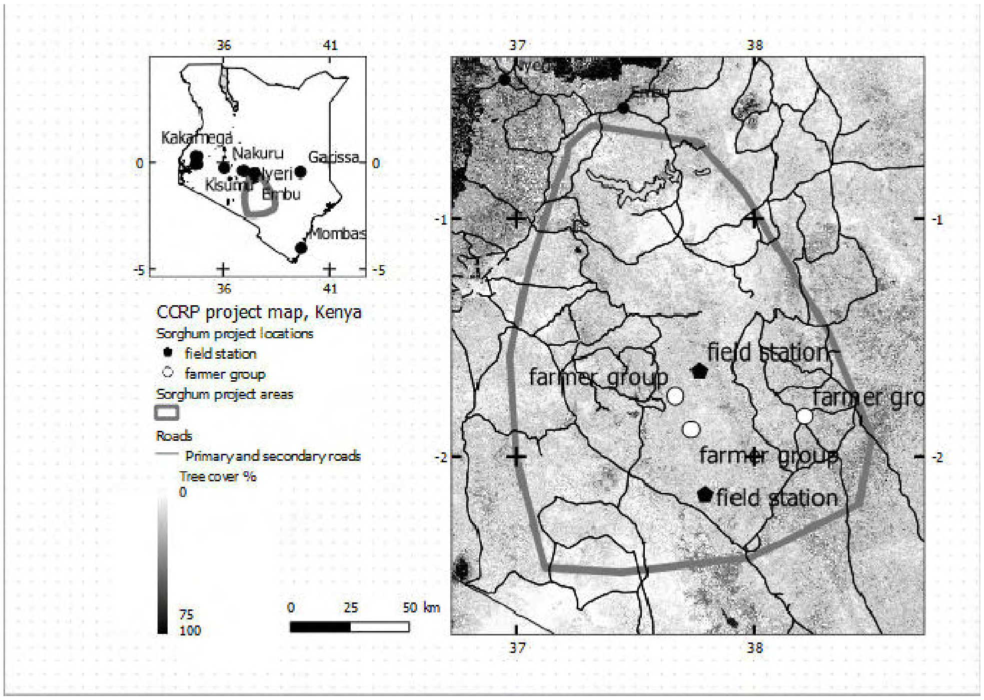


The Map unit is 1 when the scale bar is showing degrees. As 1 degree is 111km, you should change the Map unit per bar unit value to 1/111. The closest you get is by typing 0.01 (100km, not 111km), as the window does not allow more decimals.

Type km to Unit label.

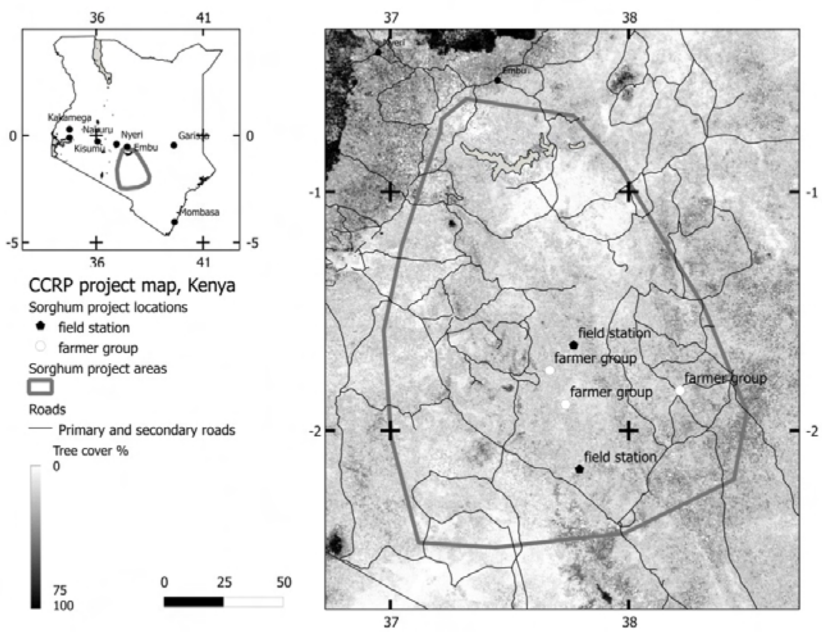
Now your scale bar is still too long for the layout. Change the Segment size to 0.25, that is, one segment is 25km.

Drag the scale bar in a suitable place in your layout and you might see something like this as your last image.



Save as Template and Export as Image (use jpeg or jpg, tiff makes very large file size).

The exported image (below) may look a bit different (e.g. text size may differ from that in the original layout).



Congratulations for following this far!